



JKBOSE 2026

CLASS 11TH

STRAIGHT LINES

MATHS

LECTURE-06

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TOPICS TO BE COVERED

1

Distance of a Point From a Line

2

Distance between Parallel Lines

3

Related Problems

Distance of a Point From a Line

Suppose that $P(x_1, y_1)$ be the given point & $ax+by+c=0$ — ①
be the given straight line.

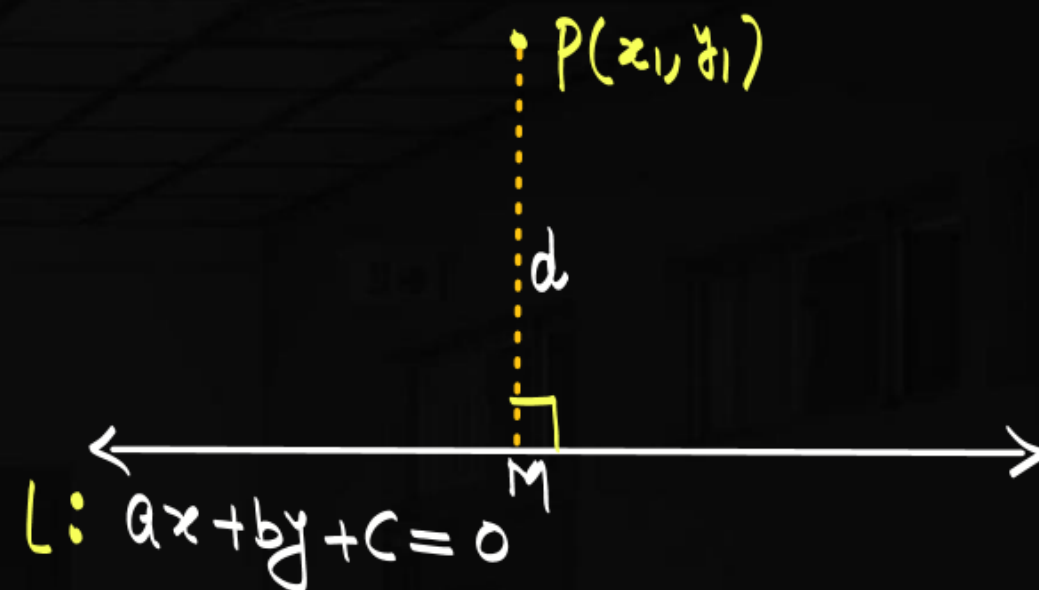
Then the distance of $P(x_1, y_1)$ from ① is given by;

$$d = PM = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

* **Question:** Find the distance of a point $P(1, 2)$ from the line $3x + 4y + 5 = 0$?

(Sol) let d be the required distance, then

$$d = \frac{|3(1) + 4(2) + 5|}{\sqrt{3^2 + 4^2}} = \frac{|3 + 8 + 5|}{\sqrt{25}}$$



$$\Rightarrow d = \frac{|16|}{5}$$

$$\Rightarrow d = \frac{16}{5} \text{ units.}$$

Hence the result.

EXAMPLE

Find the coordinates of the point on $x + y + 3 = 0$, whose distance from $x + 2y + 2 = 0$ is $\sqrt{5}$.

(Sol) let $P(h, k)$ be the required point on

$x + y + 3 = 0$ whose distance from $x + 2y + 2 = 0$ is $\sqrt{5}$.

i.e. $PM = \sqrt{5}$.

$$\Rightarrow \frac{|h + 2k + 2|}{\sqrt{1 + 4}} = \sqrt{5}$$

$$\Rightarrow |h + 2k + 2| = \sqrt{5} \times \sqrt{5}$$

$$\Rightarrow |h + 2k + 2| = 5$$

$$\Rightarrow h + 2k + 2 = \pm 5$$

Case I: $h + 2k + 2 = 5$

$$\Rightarrow h + 2k = 3 \quad \text{--- (1)}$$

Since $P(h, k)$ lies on $x + y + 3 = 0$, so

$$h + k + 3 = 0 \Rightarrow h + k = -3 \quad \text{--- (2)}$$

Subtracting (1) & (2), we get:

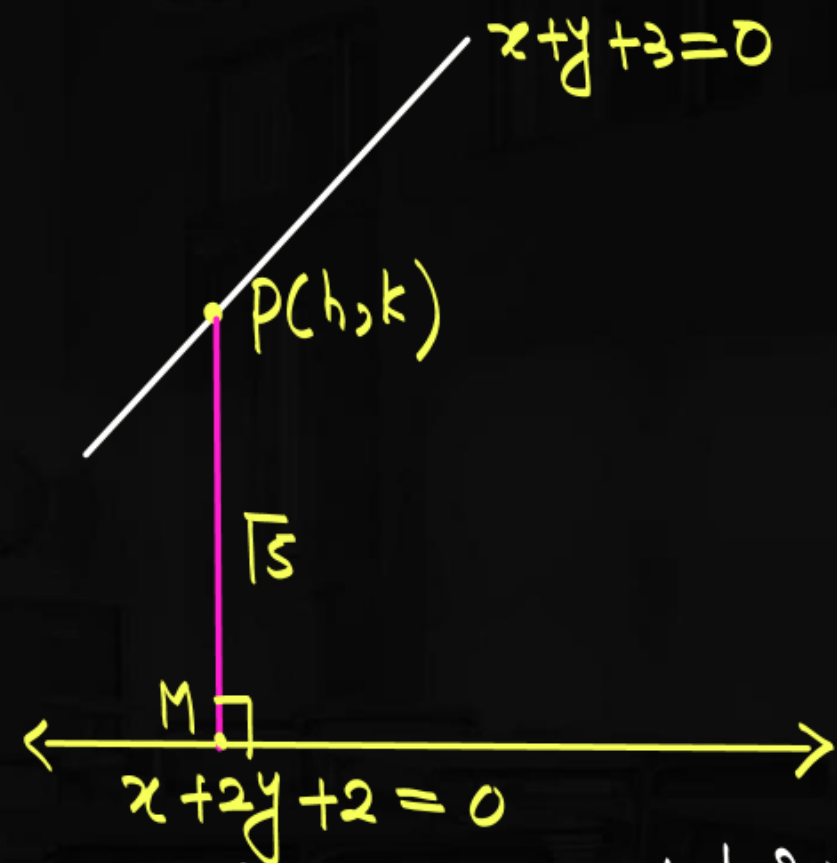
$$h + 2k - h - k = 3 + 3$$

$$\Rightarrow k = 6$$

$$\text{So, } h + 12 = 3$$

$$\Rightarrow h = -9$$

So, $(-9, 6)$ is the required point in this case.



Case 2nd: $h + 2k + 2 = -5$

$\Rightarrow h + 2k = -7$ — (3)

Since $P(h, k)$ lies on $x + y + 3 = 0$, so

$h + k + 3 = 0 \Rightarrow h + k = -3$ — (4)

Subtracting (3) & (4), we get:

$\cancel{h} + 2k - \cancel{h} - k = -7 + 3$

$\Rightarrow k = -4$.

So, $h - 8 = -7$.

$\Rightarrow h = -7 + 8$

$\Rightarrow h = 1$.

so $P(1, -4)$ is the required point in this case.

Hence $(1, -4)$ & $(-9, 6)$ are the required points.

Hence the result.

QUESTION (NCERT EXAMPLAR):

The distance of the point $P(1, -3)$ from the line $2y - 3x = 4$ is

1 13 ✗

2 $\frac{7}{13}\sqrt{13}$ ✗

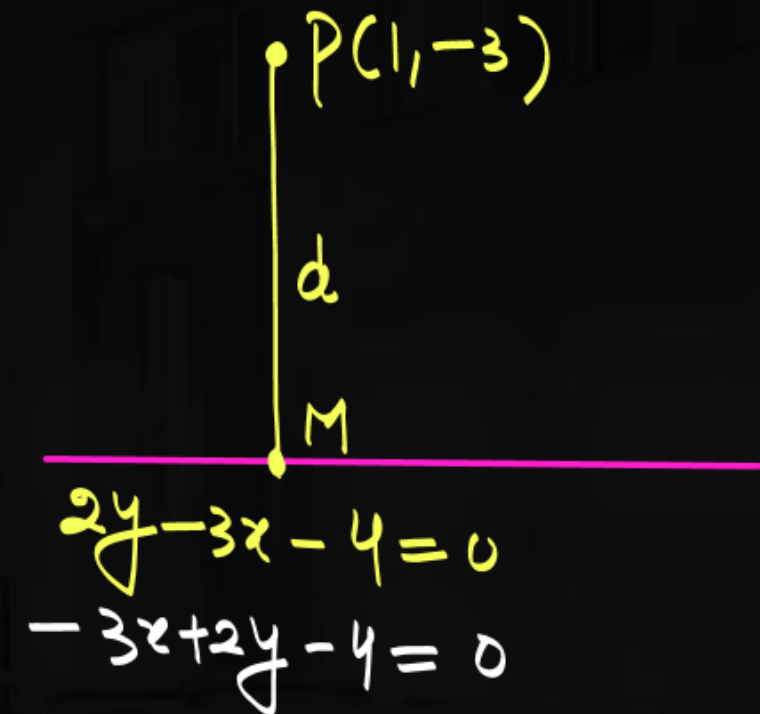
3 $\sqrt{13}$ ✓

4 None of these ✗

$$d = \frac{|-3(1) + 2(-3) - 4|}{\sqrt{9 + 4}}$$

$$\Rightarrow d = \frac{|-3 - 6 - 4|}{\sqrt{13}} = \frac{13}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}}$$

$$\Rightarrow d = \frac{13\sqrt{13}}{13} \Rightarrow d = \sqrt{13}$$



QUESTION

$$ax + by + c = 0$$

$$- \frac{a}{b}$$

$$PM = MQ$$

The image of the point $(4, -3)$ with respect to the line $y = x$ is :

1 $\times (-4, -3)$

2 $\times (3, 4)$

3 $\times (-4, 3)$

4 $\checkmark (-3, 4)$

(Sol) Since $M(\frac{h+4}{2}, \frac{k-3}{2})$ lies on $x - y = 0$, so

$$\left(\frac{h+4}{2}\right) - \left(\frac{k-3}{2}\right) = 0$$

$$\Rightarrow \frac{h+4}{2} = \frac{k-3}{2}$$

$$\Rightarrow h = k - 7 \quad \text{--- (1)}$$

Here $PQ \perp L$, so $\left(\frac{-1}{1}\right)\left(\frac{k+3}{h-4}\right) = -1$

$$\Rightarrow \frac{k+3}{h-4} = -1$$

$$\Rightarrow k+3 = -h+4$$

$$\Rightarrow k = -h+1$$

$$\Rightarrow k = -(k-7) + 1 \quad (\text{using (1)})$$

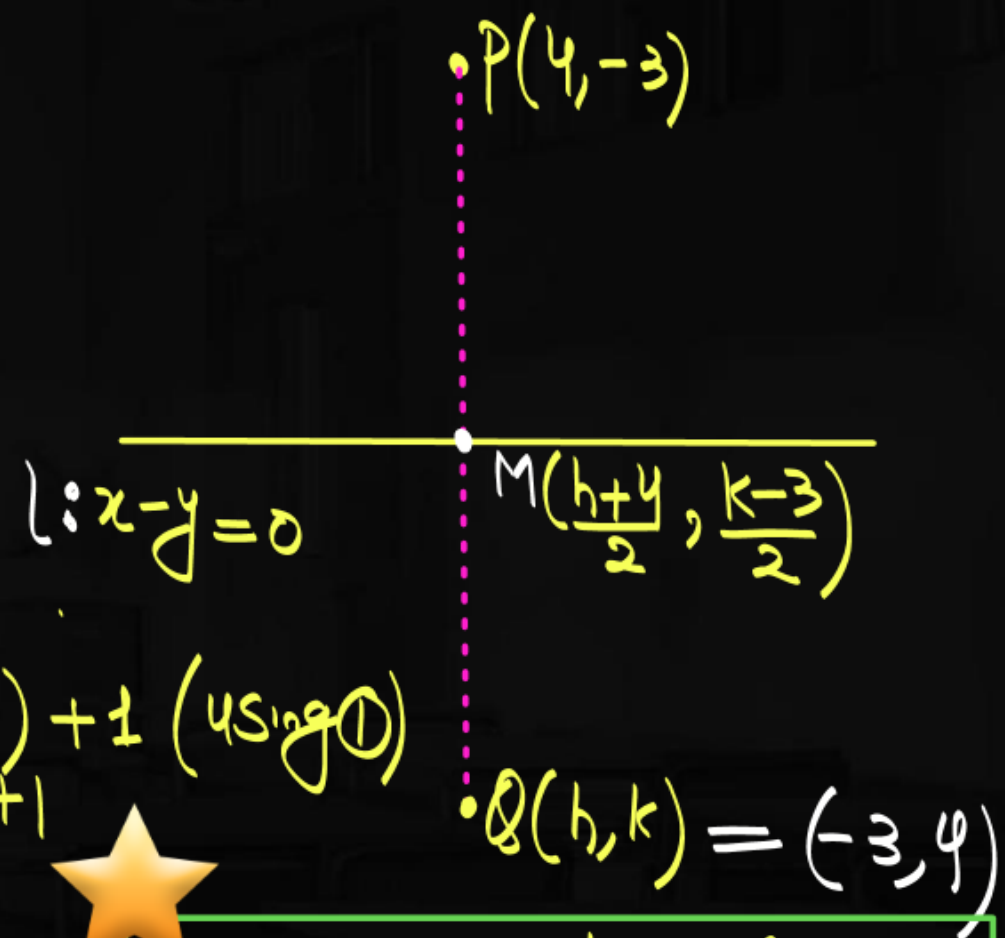
$$\Rightarrow k = -k + 7 + 1$$

$$\Rightarrow 2k = 8$$

$$\Rightarrow k = 4$$

$$\text{So, } h = 4 - 7$$

$$\Rightarrow h = -3$$



The Image of a point $P(h, k)$ w.r.t. the line $y = x$ is $Q(k, h)$

QUESTION

If $(-2, 6)$ is the **image** of the point $(4, 2)$ with respect to the line L, then the equation of L is :

$y - y_1 = m(x - x_1)$
 $PM = MQ$

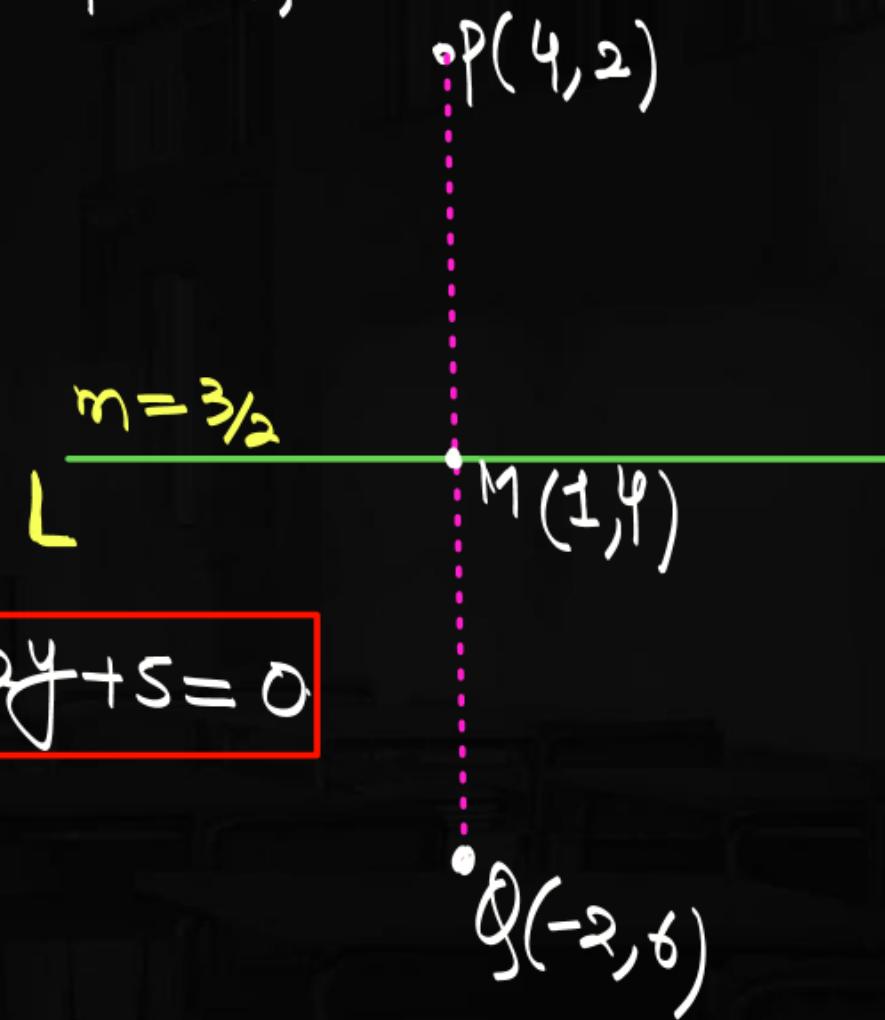
1 $3x - 2y + 10 = 0$ ~~✗~~ (Sol) let m be the slope of line L, then

2 $3x - 2y + 5 = 0$ ~~✗~~ $m \times m_{PQ} = -1$
 $\Rightarrow m \left(\frac{6-2}{-2-4} \right) = -1$

3 $6x - 4y - 7 = 0$ ~~✗~~ $\Rightarrow m \times \frac{4}{-6} = -1$
 $\Rightarrow m = \frac{6}{4} = \frac{3}{2}$

4 $2x + 3y - 5 = 0$ ~~✗~~ So the required equation is:
 $y - 4 = \frac{3}{2}(x - 1)$
 $\Rightarrow 2y - 8 = 3x - 3$

$3x - 2y + 5 = 0$



Distance between two parallel lines

Line parallel to the given line:

Suppose that $a'x + b'y + c' = 0$ — (1)
is the required line parallel to the given
line $ax + by + c = 0$ — (2)

$$\text{Then } \frac{a'}{a} = \frac{b'}{b} = \frac{c'}{c}$$

$$\Rightarrow \frac{a'}{a} = \frac{b'}{b} = \lambda$$

$$\Rightarrow a' = \lambda a \text{ \& } b' = \lambda b$$

So from (1), we get;

$$\lambda ax + \lambda by + c' = 0$$

$$\Rightarrow \lambda(ax + by) + c' = 0$$

$$ax + by + c = 0$$

$$a'x + b'y + c' = 0$$

($ax + by + k = 0$)

$$\Rightarrow ax + by + \frac{c'}{\lambda} = 0$$

$$\Rightarrow ax + by + k = 0$$

* Suppose that $ax + by + c_1 = 0$
& $ax + by + c_2 = 0$ be the given
two parallel lines. Then the distance
b/w the given parallel lines is:

$$d = \frac{|c_2 - c_1|}{\sqrt{a^2 + b^2}}$$

$$ax + by + c_1 = 0$$

d

$$ax + by + c_2 = 0$$

Question: Find the distance b/w the parallel
lines $3x + 4y + 10 = 0$ & $3x + 4y + 20 = 0$

(Sol) Here $d = \frac{|20 - 10|}{\sqrt{3^2 + 4^2}}$

$$\Rightarrow d = \frac{10}{5} = 2$$

$$\Rightarrow d = 2 \text{ units.}$$

$$3x + 4y + 10$$

d

$$3x + 4y + 20$$

QUESTION (NCERT EXAMPLAR):

The coordinates of the foot of the perpendicular from the point $(2, 3)$ on the line $x + y - 11 = 0$ are

1 ✗ $(-6, 5)$

2 ✓ $(5, 6)$

3 ✗ $(-5, 6)$

4 ✗ $(6, 5)$

(So) $M(h, k)$ lies on $x + y - 11 = 0$,

so $h + k - 11 = 0$

$\Rightarrow h = 11 - k$ — ①

Also product of slopes = -1

$\Rightarrow \left(\frac{-1}{1}\right)\left(\frac{k-3}{h-2}\right) = -1$

$\Rightarrow \frac{k-3}{h-2} = 1$

$\Rightarrow k - 3 = h - 2$

$\Rightarrow k = h + 1$

$\Rightarrow k = 11 - k + 1$. (from ①)

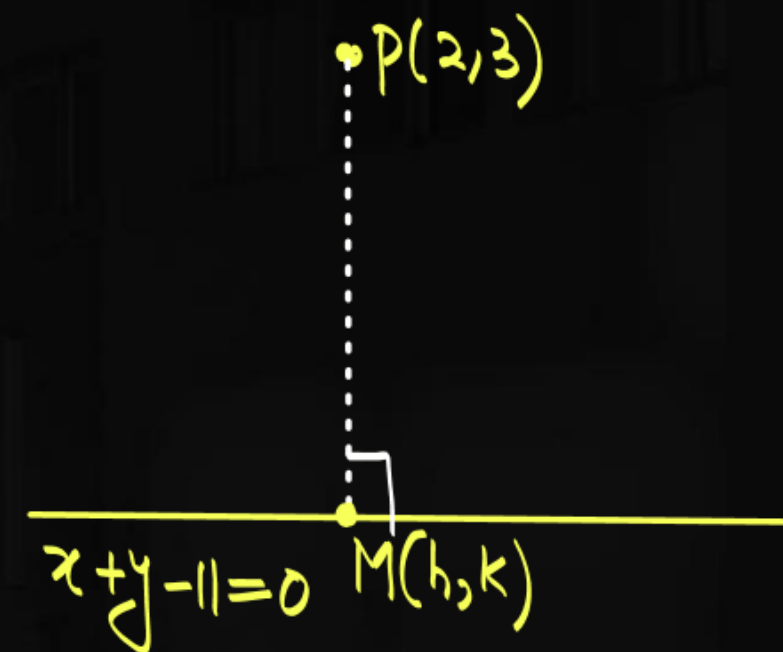
$\Rightarrow 2k = 12$

$\Rightarrow k = 6$

So, $h = 11 - 6$

$\Rightarrow h = 5$

$\Rightarrow M(h, k) = M(5, 6)$



QUESTION (NCERT EXAMPLAR):

The distance of the point of intersection of the lines $2x - 3y + 5 = 0$ and $3x + 4y = 0$ from the line $5x - 2y = 0$ is

1 $\frac{130}{17\sqrt{29}}$ *(Homework Problem)*

2 $\frac{13}{7\sqrt{29}}$

3 $\frac{130}{7}$

4 None of these

QUESTION (NCERT EXAMPLAR):

The distance between the lines $y = mx + c_1$ and $y = mx + c_2$ is

~~1~~ $\frac{c_1 - c_2}{\sqrt{m^2 + 1}}$

$$\begin{array}{c} \overleftrightarrow{mx - y + c_1 = 0} \\ | \\ \overleftrightarrow{mx - y + c_2 = 0} \end{array}$$

$$d = \frac{|c_1 - c_2|}{\sqrt{1 + m^2}}$$

~~2~~ $\frac{|c_1 - c_2|}{\sqrt{1 + m^2}}$

~~3~~ $\frac{c_2 - c_1}{\sqrt{1 + m^2}}$

~~4~~ 0

$$\begin{array}{c} \overleftrightarrow{ax + by + c' = 0} \\ | \\ \overleftrightarrow{ax + by + c = 0} \\ d = \frac{|c' - c|}{\sqrt{a^2 + b^2}} \end{array}$$

QUESTION (NCERT EXAMPLAR):

The coordinates of the foot of perpendiculars from the point $(2, 3)$ on the line $y = 3x + 4$ is given by.

1 ✗ $\left(\frac{37}{10}, \frac{-1}{10}\right)$

2 ✓ $\left(\frac{-1}{10}, \frac{37}{10}\right)$

3 ✗ $\left(\frac{37}{10}, -10\right)$

4 ✗ $\left(\frac{2}{3}, -\frac{1}{3}\right)$

(Sol) * $k = 3h + 4$ — (1)
 $3\left(\frac{k-3}{h-2}\right) = -1.$

$\Rightarrow 3k - 9 = -h + 2$

$\Rightarrow 3(3h + 4) + h = 11.$

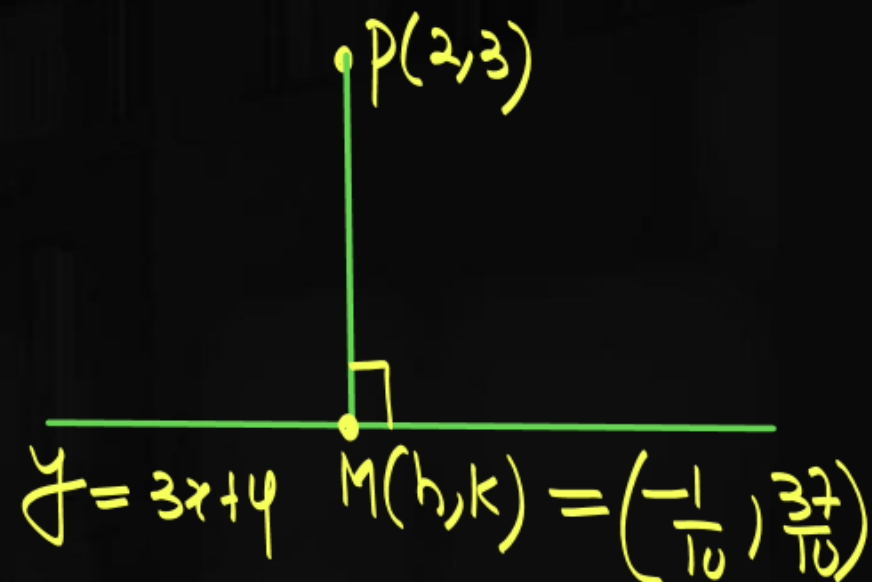
$\Rightarrow 9h + 12 + h = 11.$

$\Rightarrow 10h = -1.$

$\Rightarrow h = -\frac{1}{10}.$

$k = 3\left(-\frac{1}{10}\right) + 4$

$k = -\frac{3}{10} + 4 = \frac{37}{10}$



QUESTION

One diagonal of a square is $3x - 4y + 8 = 0$ and one vertex is $(-1, 1)$, then the area of square is : (Homework Problem)

1 $\frac{1}{50}$ sq.unit ✗

2 $\frac{1}{25}$ sq.unit ✗

3 $\frac{3}{50}$ sq.unit ✗

4 $\frac{2}{25}$ sq.unit ✓

$$d = \frac{|3(-1) - 4(1) + 8|}{\sqrt{3^2 + 4^2}}$$

$$\Rightarrow d = \frac{1}{5}$$

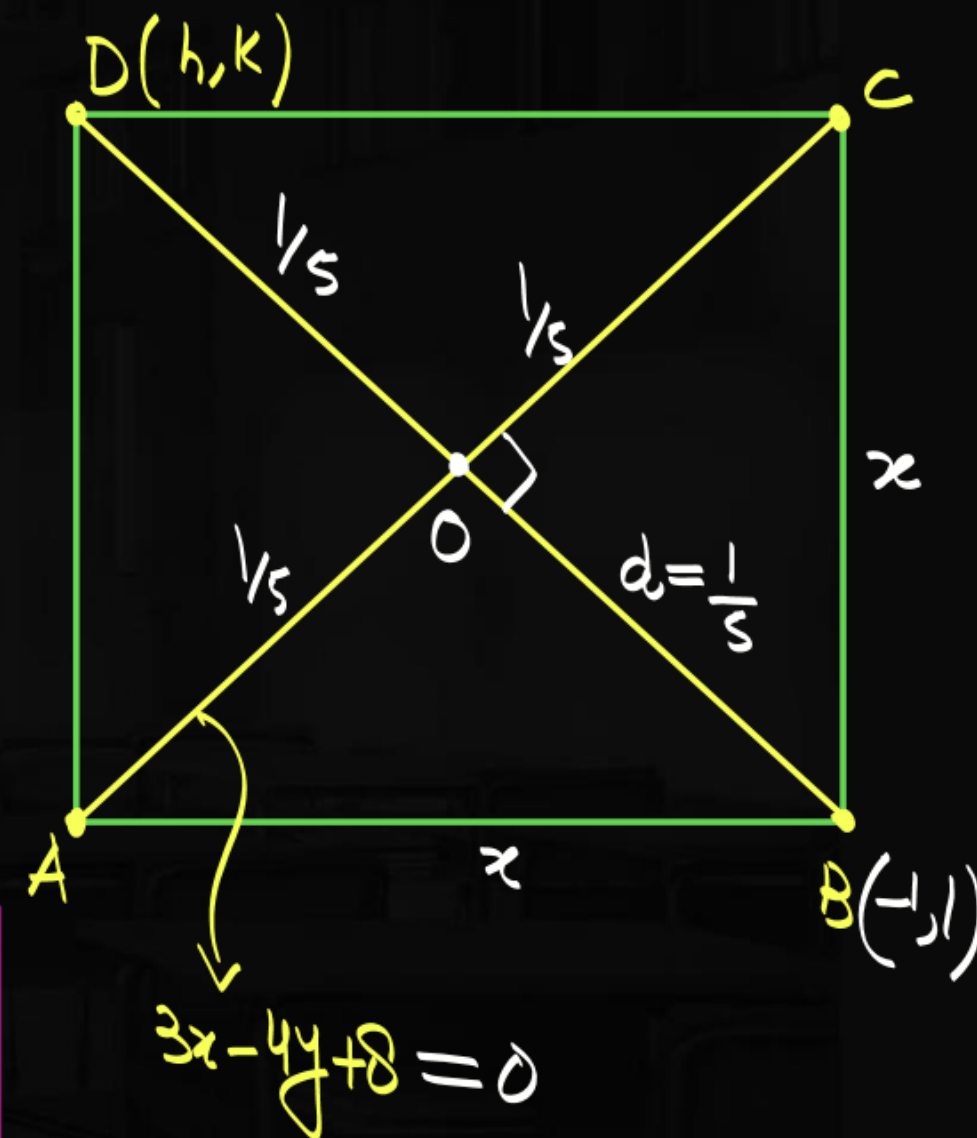
$$x^2 + x^2 = \left(\frac{2}{5}\right)^2$$

$$\Rightarrow 2x^2 = \frac{4}{25}$$

$$\Rightarrow x^2 = \frac{2}{25}$$

Area of Square
 $= \frac{2}{25}$ Sq. units.

* Homework Problem:
 Find the Co-ordinates of point D





$$\frac{|C_1 - C_2|}{\sqrt{m^2 + (-1)^2}} = \frac{|C_1 - C_2|}{\sqrt{1 + m^2}}$$

Thank You

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Rather

